

2. Top K Frequent Elements

Given an integer array `nums` and an integer `k`, return *the k most frequent elements*. You may return the answer in **any order**.

Example 1:

Input: `nums = [1,1,1,2,2,3]`, `k = 2`

Output: `[1,2]`

Example 2:

Input: `nums = [1]`, `k = 1`

Output: `[1]`

Example 3:

Input: `nums = [1,2,1,2,1,2,3,1,3,2]`, `k = 2`

Output: `[1,2]`

Constraints:

- $1 \leq \text{nums.length} \leq 10^5$
- $-10^4 \leq \text{nums}[i] \leq 10^4$
- `k` is in the range `[1, the number of unique elements in the array]`.
- It is **guaranteed** that the answer is **unique**.

PROGRAM

```
import java.util.*;
```

```
class Solution {
```

```
    public int[] topKFrequent(int[] nums, int k) {
```

```
        // Count frequency of each number
```

```
        Map<Integer, Integer> map = new HashMap<>();
```

```
        for (int num : nums) {
```

```
            map.put(num, map.getOrDefault(num, 0) + 1);
```

```
        }
```

```
        // Max heap based on frequency
```

```
        PriorityQueue<Integer> pq = new PriorityQueue<>()
```

```
(a, b) -> map.get(b) - map.get(a)

);

// Add all unique numbers
pq.addAll(map.keySet());

// Store k most frequent elements
int[] result = new int[k];

for (int i = 0; i < k; i++) {
    result[i] = pq.poll();
}

return result;
}
}
```

OUTPUT

github - Searchvtu30221-cmyk/PST-programsRemove Duplicates In PlaceTop K Frequent Elements - LeetCode

https://leetcode.com/problems/top-k-frequent-elements/submissions/2145304711/

Problem List<>>Submit

Accepted25 / 25 testcases passedVTU30221 submitted at Sep 18, 2026 09:07AnalysisSolution

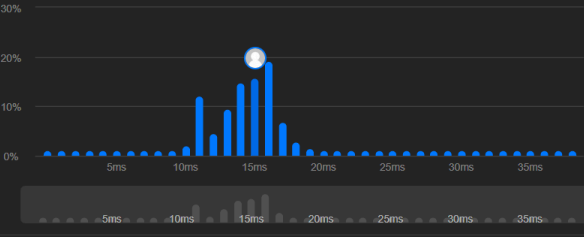
School's in. Leet's on.
Get LeetCode Annual Premium for \$99!

Runtime

15 msBeat: 55.00%

Memory

47.46 MBBeat: 81.90%



Code | Java

1 import java.util.*;

Code

JavaAuto

```
22 int[] result = new int[k];
23
24 for (int i = 0; i < k; i++) {
25     result[i] = pq.poll();
26 }
27
28 return result;
29 }
30 }
31
```

SavedLn 31, Col 1

TestcaseTest Result

AcceptedRuntime: 2 ms

Case 1Case 2Case 3

Input

nums =
[1,1,1,2,2,3]

k =
2

Output

30°C
Mostly sunny

Search

ENG
IN

73%

09:07
18-09-2026